

1.
a)

$$f(x, y) = \frac{x}{8} - \frac{y}{5} - 1 = 0$$

in x intercept, $y = 0$

$$\therefore \frac{x}{8} - \frac{0}{5} - 1 = 0$$

$$\Rightarrow \frac{x}{8} - 0 - 1 = 0$$

$$\Rightarrow \frac{x}{8} = 1$$

$$\Rightarrow x = 8$$

start (8, 0)

in y intercept, $x = 0$

$$\therefore \frac{0}{8} - \frac{y}{5} - 1 = 0$$

$$0 - \frac{y}{5} - 1 = 0$$

$$\Rightarrow -\frac{y}{5} = 1$$

$$\Rightarrow y = -5$$

End (0, -5)

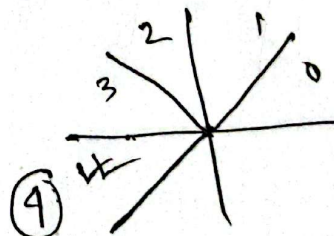
start (8, 0)

$$\therefore dx = 0 - 8 = -8$$

$$dy = -5 - 0 = -5$$

$$|dx| > |dy|$$

Zone $\rightarrow$ 4



b) in Zone -4

start (8, 0)
End (0, -5)

convert to
Zone 0

Zone $\rightarrow 0$

$$x' = -x$$

$$y' = -y \therefore (-x, -y)$$

start (-8, 0)

End (0, 5)

$$\therefore dy = 5 - 0 = 5$$

$$dx = 0 - 8 = -8$$

$$\therefore D = 2dy - dx = 2 \times 5 - 8 = 10 - 8 = 2$$

$$\Rightarrow D = 2$$

$$\Delta NE = 2(dy - dx) = 2(5 - 8) = 2(-3) = -6$$

$$\Delta NE = -6$$

$$\therefore \Delta E = 2dy = 2 \times 5 = 10$$

$$\Delta E = 10$$

				Zone 4	
x'	y'	D	$\Delta NE / \Delta E$	$x = -x'$	$y = -y'$
-8	0	2	NE	8	0
-7	1	-4	E	7	-1
-6	1	6	NE	6	-1
-5	2	0	E	5	-2
-4	2	10	NE	4	-2

$\therefore$ pixels: (8, 0)
(7, -1)
(6, -1)
(5, -2)
(4, -2)